Week 4 · Part A — diagnoses of the nine charts

Written for: the marker of CDB601220 Week 4. Each case: the conclusion the chart invites, the decoy objection and why it fails, the mechanism with its number, and the redesign (figure: left = rebuilt as published, right = redesign). All numbers are printed by partA_charts.py (partA_output.txt). Data is synthetic course data.

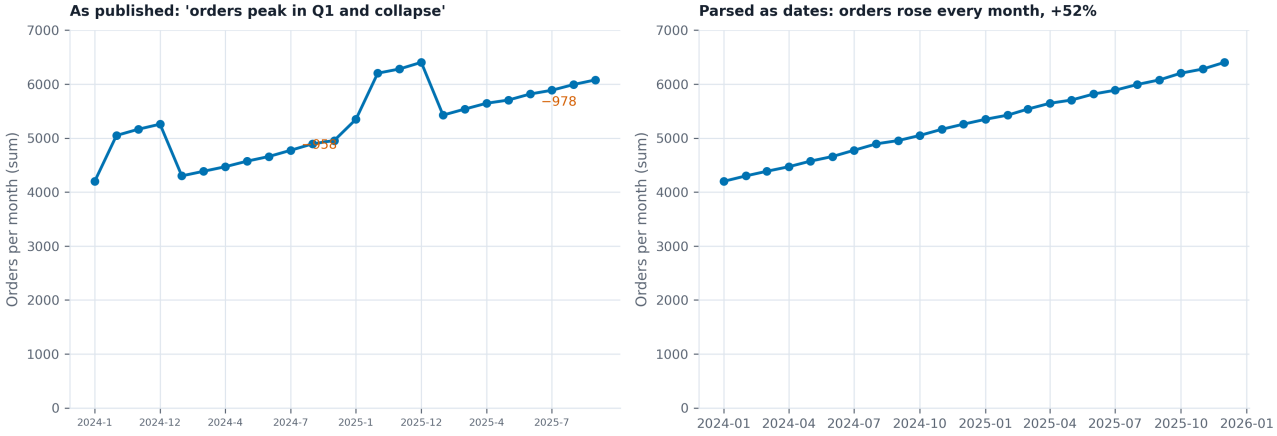
1 · “Orders peak in Q1 and collapse every year”

Conclusion invited: A strong annual cycle: orders climb Jan–Apr, fall in spring, repeat.

Decoy: 'real seasonality'. Ruled out: the true series (Jan 2024 4,200 → Dec 2025 6,405, +52.5%) is monotone; its worst month-on-month change is +59, never negative.

Mechanism: *month* is text written YYYY-M with no padding, so a text sort puts '2024-10','2024-11','2024-12' after '2024-1' and before '2024-2'. 22 of 24 rows land in the wrong place; the two 'collapses' (–958, –978) are the jumps from Dec back to Feb.

Redesign: parse with format='%Y-%m', sort as datetime — one unbroken rise.



Redesign: month parsed with format='%Y-%m' and sorted as datetime. Synthetic course data, CDB601220 Week 4.

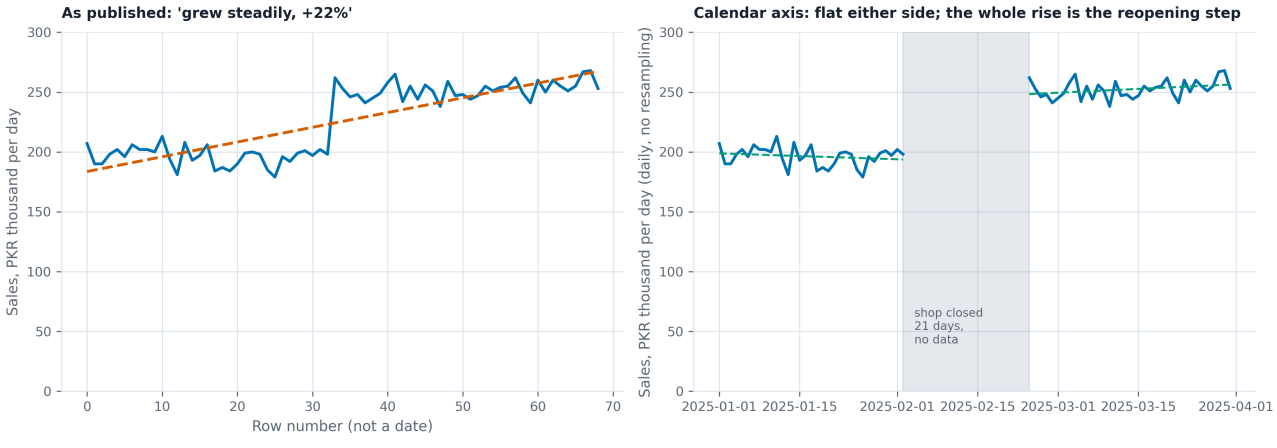
2 · “Sales grew steadily, 22%, across the quarter”

Conclusion invited: A steady climb 207 → 253 (+22%) with a rising trend line.

Decoy: 'the dates are just labels'. Ruled out: 69 rows span 90 calendar days — 21 days never trade (3 Feb – 23 Feb).

Mechanism: plotting by row number deletes the 21-day closure. Before it sales fall 207 → 198 (–0.16k/day); after it they run 262 → 253 (+0.23k/day): flat each side. The whole rise is a single +64k step across the closure; a trend line fitted across it (+1.23/row) measures the step, not growth.

Redesign: calendar axis, closure left as a gap, one trend per trading spell.



Redesign: calendar axis, gap left as NaN, separate trend per trading spell. Synthetic course data, CDB601220 Week 4.

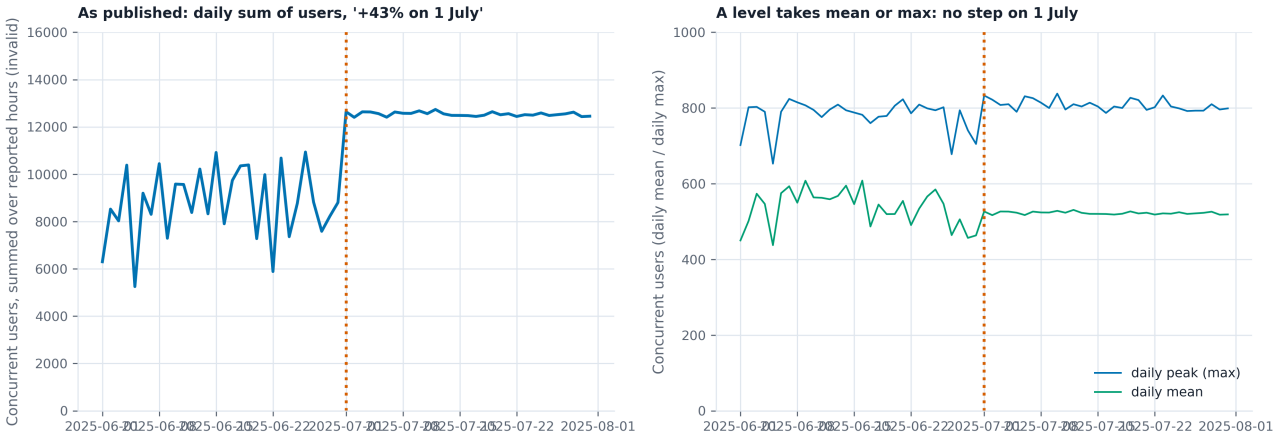
3 · “Peak load jumped 43% overnight on 1 July”

Conclusion invited: A capacity step on 1 July (daily June mean 8,782 → July 12,543, +42.8%).

Decoy: 'July is just busier'. Ruled out: daily mean users go 535.9 → 522.6 (–2.5%) and daily max 779 → 808 (+3.7%).

Mechanism: concurrent_users is a *level* (users connected at an instant); summing hourly levels counts coverage, not load. June has only 493 of 720 hours recorded (16.4 h/day), July 744 of 744 (24 h/day); 24/16.4 ≈ 1.46 reproduces the 'jump'. Bytes served (a flow) also 'jump' +42.7% for the same reason, but per recorded hour fall –2.3%.

Redesign: a level takes mean or max per day — no step on 1 July.



Redesign: level aggregated with mean and max over the hours actually recorded. Synthetic course data, CDB601220 Week 4.

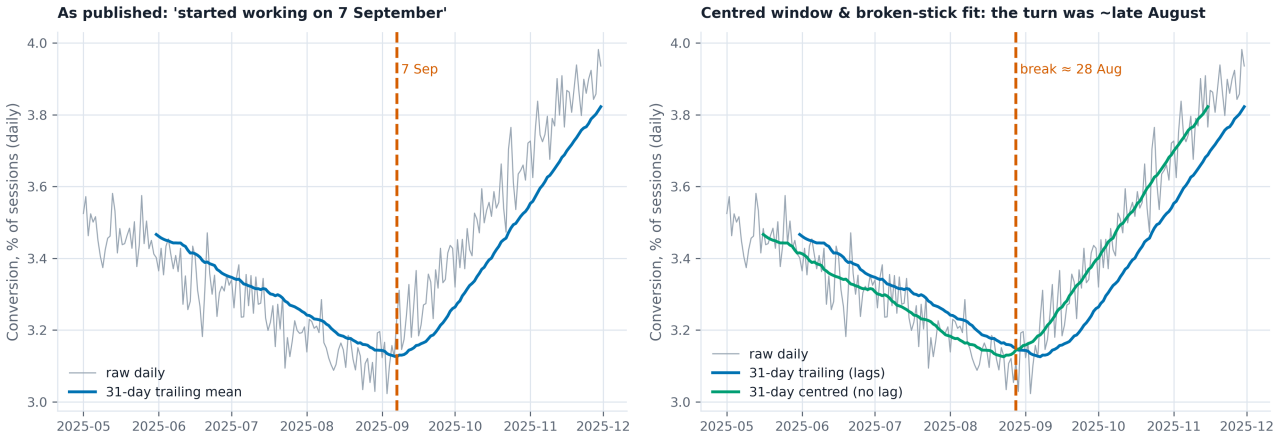
4 · “The new checkout flow started working on 7 September”

Conclusion invited: A trailing 31-day mean bottoms on 2025-09-07, so that is when the fix kicked in.

Decoy: 'the fix has a delayed effect'. Ruled out: the raw data itself turn earlier.

Mechanism: a trailing mean at date t averages days t–30…t, so it cannot register a turn until the window has filled: it is half a window late by arithmetic. Centred-31 minimum: 2025-08-23; a broken-stick fit on the raw data puts the break at 2025-08-28; lag_at_turn gives 15 days = the half-window (15).

Redesign: show the centred window and the fitted break; caption the 31-day window.



Window length 31 days named on both panels; centred version for diagnosis only. Synthetic course data, CDB601220 Week 4.

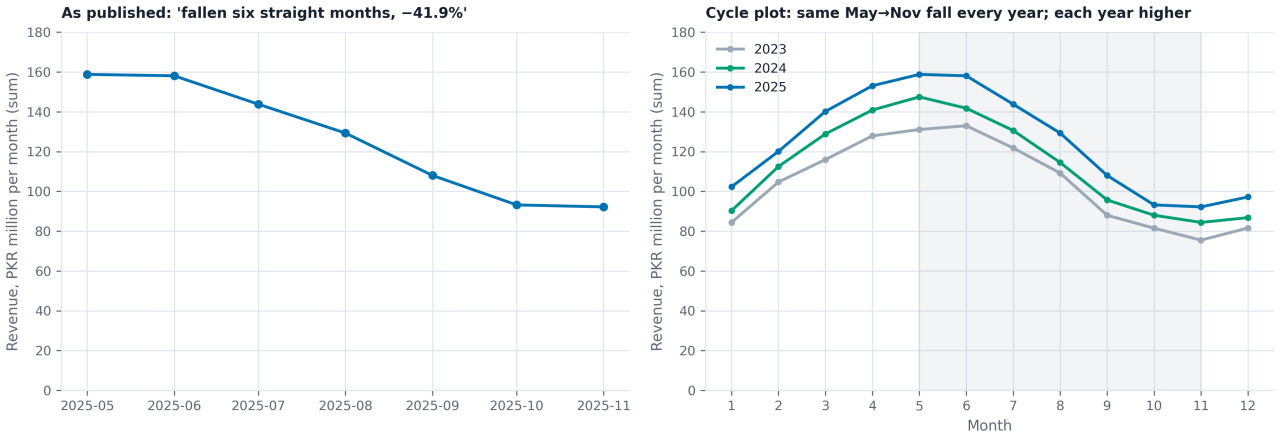
5 · “Revenue has fallen for six straight months”

Conclusion invited: A sustained collapse, 158.8 → 92.2 (–41.9%).

Decoy: truncated axis / tiny sample. Ruled out: the recipe starts y at zero and every point is a full month of the 36.

Mechanism: the window starts at the seasonal peak (May) and ends at the trough (Nov). May → Nov is –42.4%, –42.8%, –41.9% in 2023, 2024, 2025: the same fall every year. Year-on-year, all 24 months of 2024–25 are positive (mean +9.2%, range +4.9% to +13.3%).

Redesign: cycle plot, each year on a Jan–Dec axis.



Redesign: each year overlaid on a Jan–Dec axis (cycle plot), y from zero. Synthetic course data, CDB601220 Week 4.

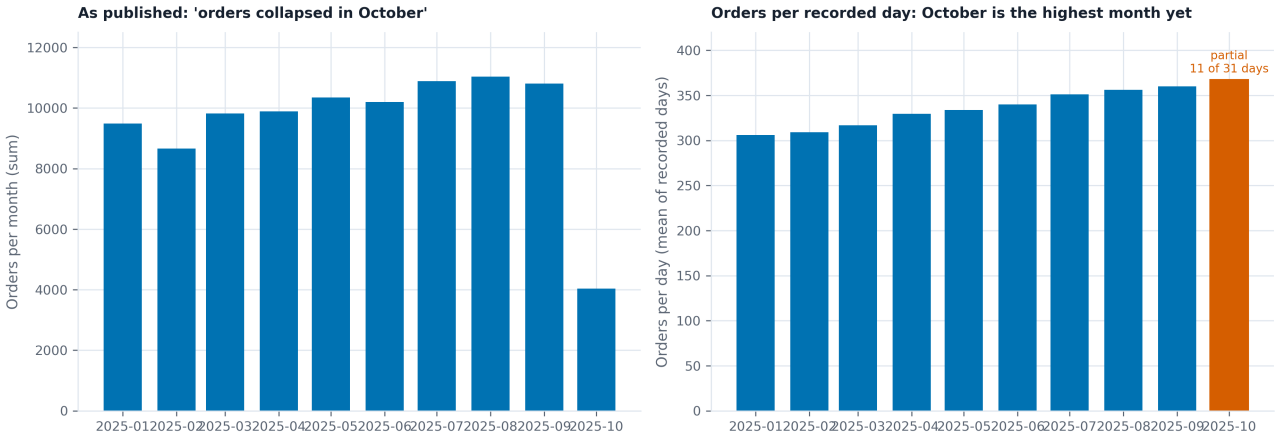
6 · “Orders collapsed in October”

Conclusion invited: October fell to 4,041 from 10,805 — 37% of September.

Decoy: 'a real demand drop'. Ruled out: the daily mean is up.

Mechanism: the extract stops 2025-10-11, so October holds 11 of 31 days. Orders per recorded day: 360.2 (Sep) → 367.4 (Oct, +2.0%), the highest of the year.

Redesign: orders per recorded day, partial month hatched and labelled.



Redesign: rate per recorded day; partial month hatched and labelled. Synthetic course data, CDB601220 Week 4.

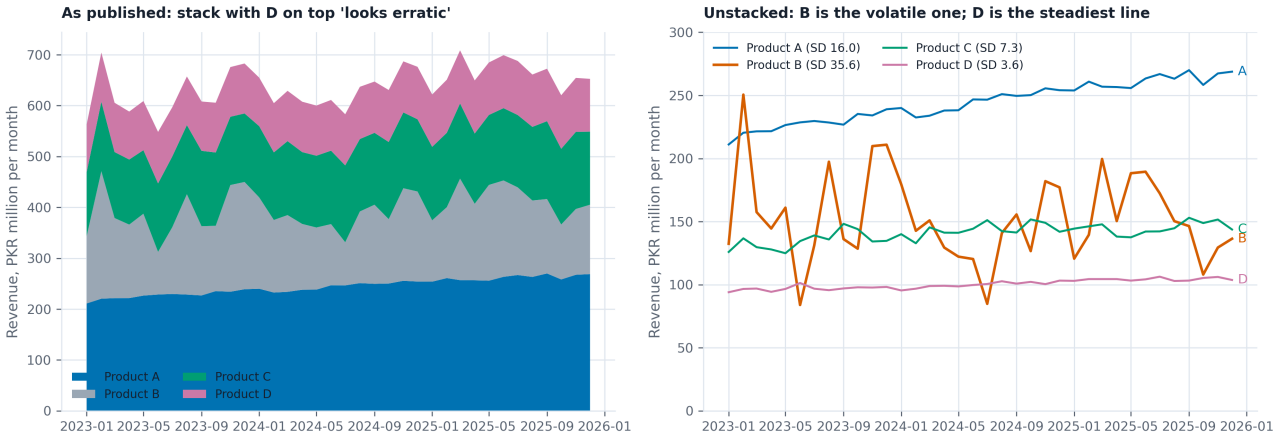
7 · “Product D has been erratic since mid-2024”

Conclusion invited: D's wobbling top edge says D is unstable.

Decoy: 'D really is volatile'. Ruled out: D's own level SD is 3.6 and month-on-month SD 2.0 — the calmest of the four.

Mechanism: a stacked area draws each layer on a moving baseline. D sits on A+B+C; B (level SD 35.6, month-on-month 45.8) is the volatile one and its swings are passed up through C into D's top edge (SD 47.3). The envelope is 548–709 because of B.

Redesign: unstack — four lines on one zero-based axis.



Redesign: lines on one shared zero-based axis (unstack). Synthetic course data, CDB601220 Week 4.

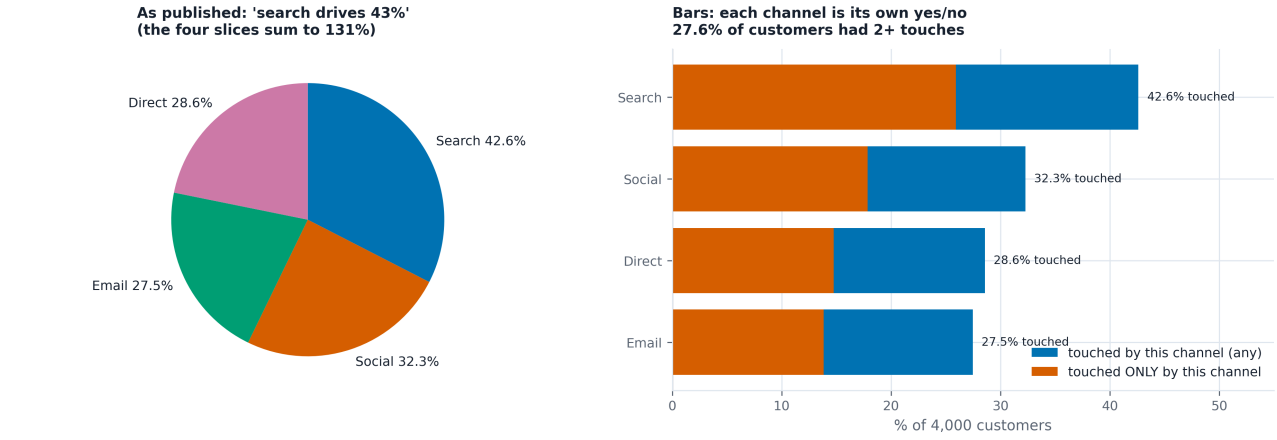
8 · “Paid search drives 43% of our customers”

Conclusion invited: Search is the dominant channel — 42.6% of the pie.

Decoy: 'rounding / unusual sample'. Ruled out: shares are exact per column.

Mechanism: the four touch flags are not exclusive: 42.6 + 32.3 + 27.5 + 28.6 = 131.0%. 1,105 of 4,000 customers (27.6%) were touched by 2+ channels (980 by two, 116 by three, 9 by four). Search is 'touched' by 42.6% but touched *only* by search for 25.9%. 'Drives' is a claim about causation the data never makes.

Redesign: one bar per channel against all customers, with the 'only this channel' overlap shown.



Redesign: one bar per channel against all customers; shares not summed. Synthetic course data, CDB601220 Week 4.

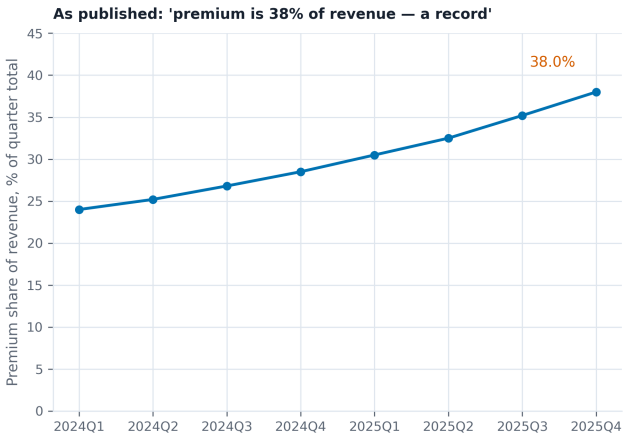
9 · “Premium is now 38% of revenue — a record”

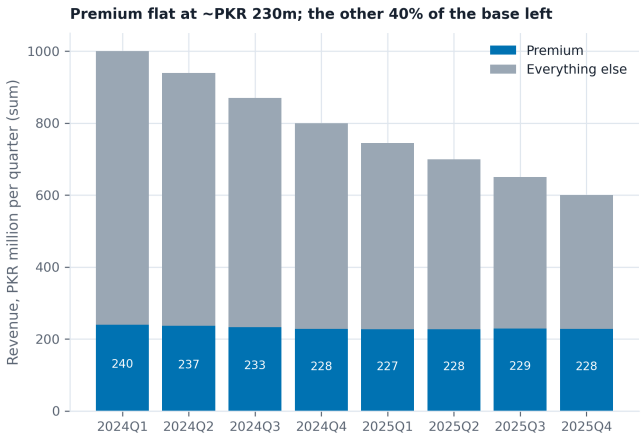
Conclusion invited: Premium is booming: 24.0% → 38.0%.

Decoy: 'premium grew'. Ruled out: premium revenue is 240 → 228 (–5.0%).

Mechanism: the denominator fell. Total revenue 1,000 → 600 (–40.0%); non-premium revenue 760 → 372 (–51.1%). A share can rise while its numerator stalls. This is incomplete rather than wrong: the share is correct, it hides the story.

Redesign: the same share, with the two absolute series beside it (stacked, zero baseline) — not a dual axis.





Redesign: the two absolute quantities behind the share, stacked, zero baseline. Synthetic course data, CDB601220 Week 4.